VR Post-experiment Questionnaires

* Required

Section 1 - Participant Information

1. Participant ID *

2. Gender *

- Female
- Male

3. Age *

4. What is your nationality? *

5. Highest education level achieved *

- High school or equivalent
- Associate degree or equivalent
- Bachelor's degree or equivalent
- Master's degree or equivalent
- Doctoral degree or equivalent

6. Dominant hand in daily activities *

- Right hand
- Left hand

7. How often do you walk to get to places? *

- Every day (7 days)
- 4 to 5 days per week
- 2 to 3 days per week
- 1 day per week
- Rarely (less than 1 day per week)
- Never

8. How often do you use VR? *

- ◯ Never
- ◯ Seldom
- ◯ Sometimes
- ◯ Often
- ◯ Very often

Automated shuttles are small, self-driving vehicles designed for passenger transportation, typically in short-distance or shared-space environments, and commonly deployed for "last-mile" journeys. These vehicles operate without a human driver, relying on advanced technologies like sensors, cameras, lidar, and machine learning to navigate and interact safely with their surroundings.

9. Are you familiar with the concept of automated shuttles? *

- ◯ Not at all
- ◯ Slightly familiar
- ◯ Moderate familiar
- ◯ Very familiar
- ◯ Extremely familiar

10. How much previous experience do you have interacting with automated shuttles? *

- ◯ No experience
- ◯ Little experience
- ◯ Some experience
- ◯ A lot of experience
- ◯ Extensive experience

Section 2 - Simulator Sickness Questionnaire

11. Did you experience any of the following symptoms during the experiment? *

	None	Slight	Moderate	Severe
General discomfort	◯	◯	◯	◯
Fatigue	◯	◯	◯	◯
Headache	◯	◯	◯	◯
Eye strain	◯	◯	◯	◯
Difficulty focusing	◯	◯	◯	◯
Increased salivation	◯	◯	◯	◯
Sweating	◯	◯	◯	◯
Nausea	◯	◯	◯	◯
Difficulty concentrating	◯	◯	◯	◯
Fullness of head	◯	◯	◯	◯
Blurred vision	◯	◯	◯	◯
Dizzy (eyes open)	◯	◯	◯	◯
Dizzy (eyes closed)	◯	◯	◯	◯
Vertigo	◯	◯	◯	◯
Stomach awareness	◯	◯	◯	◯
Burping	◯	◯	◯	◯

Section 3 - Presence Questionnaire

12. How much were you able to control events (e.g., control your movement or actions in the virtual environment)? *

Not at all	Very little	Slightly	Somewhat	Quite a bit	Mostly	Completely
○	○	○	○	○	○	○

13. How responsive was the environment to actions that you initiated or performed? *

Not responsive	Slightly responsive	Somewhat responsive	Moderately responsive	Quite responsive	Very responsive	Completely responsive
○	○	○	○	○	○	○

14. How natural did your interactions with the environment seem? *

Extremely artificial	Very artificial	Somewhat artificial	Borderline	Somewhat natural	Very natural	Completely natural
○	○	○	○	○	○	○

15. How much did the visual aspects (e.g., automated shuttles, surroundings) of the environment involve you? *

Not at all	Very little	Slightly	Somewhat	Quite a bit	A lot	Completely
○	○	○	○	○	○	○

16. How natural was the mechanism that controlled movement through the environment (i.e., physically walking)? *

Extremely artificial	Very artificial	Somewhat artificial	Borderline	Somewhat natural	Very natural	Completely natural
○	○	○	○	○	○	○

17. How compelling or convincing was your sense of objects moving through space? *

Not compelling	Slightly compelling	Somewhat compelling	Moderately compelling	Fairly compelling	Quite compelling	Very compelling
○	○	○	○	○	○	○

18. How much did your experiences in the virtual environment seem consistent with your real world experiences? *

Not consistent	Slightly consistent	Somewhat c onsistent	Moderately consistent	Fairly consistent	Quite consist ent	Very consist ent
○	○	○	○	○	○	○

19. How completely were you able to actively survey or search the environment using vision? *

Not at all	Very little	Slightly	Somewhat	Moderately	Very much	Completely
○	○	○	○	○	○	○

20. How compelling or convincing was your sense of moving around inside the virtual environment? *

Not compelling	Slightly compelling	Somewhat compelling	Moderately compelling	Fairly compelling	Quite compelling	Very compelling
○	○	○	○	○	○	○

21. How well could you move or manipulate objects in the virtual environment? *

Not at all	Very poorly	Poorly	Somewhat	Well	Very well	Extensively
○	○	○	○	○	○	○

22. How involved were you in the virtual environment experience? *

Not involved	Barely involved	Slightly involved	Mildly involved	Moderately involved	Very involved	Completely involved
○	○	○	○	○	○	○

23. How easy was it to identify objects through physical interaction, like touching an object, walking over a surface, or bumping into a wall or object? *

Not at all	Very difficult	Somewhat difficult	Neutral	Somewhat easy	Very easy	Completely easy
○	○	○	○	○	○	○

24. How much did the auditory aspects (i.e., sounds from the surroundings and automated shuttles) of the environment involve you? *

Not at all	Very little	Slightly	Somewhat	Quite a bit	A lot	Completely
○	○	○	○	○	○	○

25. How well could you identify sounds? *

Not at all	Very poorly	Poorly	Somewhat	Well	Very well	Completely
○	○	○	○	○	○	○

26. How well could you localize sounds? *

Not at all	Very poorly	Poorly	Somewhat	Well	Very well	Completely
○	○	○	○	○	○	○

27. How well could you actively survey or search the virtual environment using touch? *

Not at all	Very poorly	Poorly	Somewhat	Well	Very well	Completely
○	○	○	○	○	○	○

28. How closely were you able to examine objects? *

Not at all	Very slightly	Slightly	Pretty closely	Fairly closely	Quite closely	Very closely
○	○	○	○	○	○	○

29. How well could you examine objects from multiple viewpoints? *

Not at all	Very poorly	Poorly	Somewhat	Well	Very well	Extensively
○	○	○	○	○	○	○

30. How much delay did you experience between your actions and expected outcomes? *

No delays	Very short delays	Short delays	Moderate delays	Somewhat long delays	Quite long delays	Long delays
○	○	○	○	○	○	○

31. How much did the visual display quality interfere or distract you from performing assigned tasks or required activities? *

Not at all	Very little	A little	Interfered somewhat	Moderately interfered	Significantly interfered	Prevented task performance
○	○	○	○	○	○	○

32. How much did the control devices (e.g., use of controllers) interfere or hinder with the performance of assigned tasks or with other activities? *

Not at all	Very little	A little	Interfered somewhat	Moderately interfered	Significantly interfered	Interfered greatly
○	○	○	○	○	○	○

33. Were you able to anticipate what would happen next in response to the actions that you performed? *

Not at all	Very poorly	Poorly	Somewhat	Well	Very well	Completely
○	○	○	○	○	○	○

34. How quickly did you adjust to the virtual environment experience? *

Not at all	Very slowly	Slowly	Somewhat slowly	Fairly quickly	Quickly	Less than 1 minute
○	○	○	○	○	○	○

35. How proficient in moving and interacting with the virtual environment did you feel at the end of the experience? *

Not proficient	Very little proficient	Slightly proficient	Reasonably proficient	Quite proficient	Proficient	Very proficient
○	○	○	○	○	○	○

36. How well could you concentrate on the assigned tasks or required activities rather than on the mechanisms used to perform those tasks or activities (e.g., the use of controllers, the way you move)? *

Not at all	Very poorly	Poorly	Somewhat	Well	Very well	Completely
○	○	○	○	○	○	○

37. How completely were your senses engaged in this experience? *

Not at all	Very little	Slightly	Somewhat	Quite a lot	A lot	Completely
○	○	○	○	○	○	○

38. Were there moments during the virtual environment experience when you felt completely focused on the task or environment? *

Not at all	Very rarely	Rarely	Somewhat	Frequently	Very frequently	Completely
○	○	○	○	○	○	○

39. How easily did you adjust to the control devices used to interact with the virtual environment? *

Not at all	Very slowly	Slowly	Somewhat slowly	Fairly quickly	Quickly	Less than 1 minute
○	○	○	○	○	○	○

40. Was the information provided through different senses in the virtual environment (e.g., vision, hearing, touch) consistent? *

Not consistent	Slightly consistent	Somewhat consistent	Moderately consistent	Fairly consistent	Quite consistent	Very consistent
○	○	○	○	○	○	○

Section 4 - Pedestrian Behavior Questionnaire

41. As a pedestrian, how often do you engage in the following behaviors **in your daily life**? *

	Very infrequently or never	Infrequently	Quite infrequently	Quite frequently	Frequently	Very frequently or always
I cross the street even though the pedestrian light is red.	○	○	○	○	○	○
I cross diagonally to save time.	○	○	○	○	○	○
I cross outside the pedestrian crossing even if there is one (crosswalk) less than 50 meters away.	○	○	○	○	○	○
I take passageways forbidden to pedestrians to save time.	○	○	○	○	○	○
I realize that I have crossed several streets and intersections without paying attention to traffic.	○	○	○	○	○	○
I forget to look before crossing because I am thinking about something else.	○	○	○	○	○	○
I cross without looking because I am talking with someone.	○	○	○	○	○	○
I forget to look before crossing because I want to join someone on the sidewalk on the other side.	○	○	○	○	○	○
I thank a driver who stops to let me cross.	○	○	○	○	○	○
When I am accompanied by other pedestrians, I walk in single file on narrow sidewalks so as not to bother the pedestrians I meet.	○	○	○	○	○	○
I walk on the default side of the sidewalk to avoid disturbing other pedestrians.	○	○	○	○	○	○
I let a car go by, even if I have the right-of-way, if there is no other vehicle behind it.	○	○	○	○	○	○

Section 5 - Trust in AV Questionnaire

42. To what extent do you agree with the following statements? *

	Strongly disagree	Disagree	Somewhat disagree	Neutral	Somewhat agree	Agree	Strongly agree
In general, I trust the automated driving system.	◯	◯	◯	◯	◯	◯	◯
In general, I trust my capacity to resume control if needed.	◯	◯	◯	◯	◯	◯	◯
I trust the automated drivi ng systems when overtaking.	◯	◯	◯	◯	◯	◯	◯
I trust the automated driving systems to keep to a lane.	◯	◯	◯	◯	◯	◯	◯
I trust the automated driving systems to avoid obstacles.	◯	◯	◯	◯	◯	◯	◯
I trust the automated driving systems to keep a distance from a vehicle ahead.	◯	◯	◯	◯	◯	◯	◯

Section 6 - Feedback

We would greatly appreciate hearing about your feelings, feedback, or suggestions regarding your VR experience as a pedestrian. Your input will help us continue to improve it.

If you don't have any feedback or if you're finished, please submit the questionnaire as this is the final section. Thank you for your interest in VR and for participating in this experiment.

43. Feedback